

Box Relations, traced to axioms

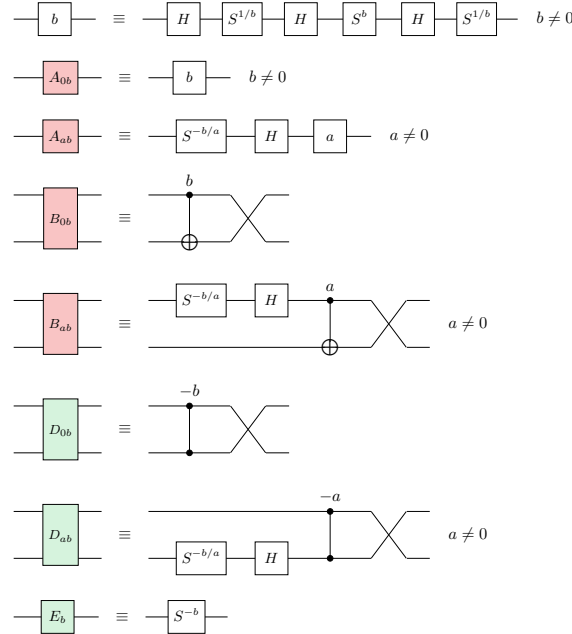
This document proves every box relation in `BoxRelations.agda` in *layers*, in the style of `test2.tex`:

- **Layer 0 (§2)** — the *axioms and core lemmas* (the leaves of every proof), each shown as a circuit rewrite.
- **Layer 1 (§3)** — each box relation: its statement, its proof chain, and a *step trace* in which every step is justified by a definitional move ($=$: box expand/fold, power combine) or by a Layer-0 lemma (tagged `[name]`).

Following a tag down to Layer 0 makes every step bottom out in an axiom. Connectives: $=$ is definitional equality; $\equiv_s$ is symplectic equivalence. Circuit order: leftmost gate is applied first; wire 0 is bottom ($\downarrow$), wire 1 is top ($\uparrow$).

1 Box definitions

Each box is a word in the generators H, S, CZ (and the derived $M[a], CX, \text{Ex}$). Folding/unfolding a box is a definitional ($=$) step.

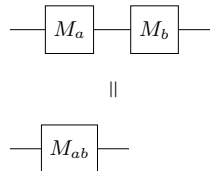


2 Layer 0 — Axioms and core lemmas

Every $\equiv_s$ step in §3 is one of the following.

[M-mul] $M[a] \cdot M[b] = M[ab]$

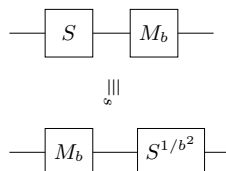
(axiom M-mul)



[semi-MS] $S \cdot M[b] \equiv_s M[b] \cdot S^{1/b^2}$

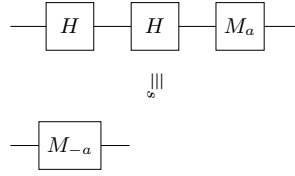
(axiom semi-MS)

S pushed through a metaplectic M box rescales by $1/b^2$.



[HHM] $H^2 \cdot M[a] \equiv_s M[-a]$

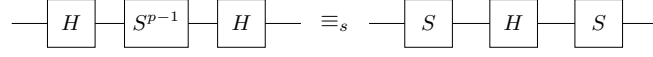
($H^2 = M[-1]$, then **[M-mul]**)



[Hconj] $H \cdot S^{-1} \cdot H \equiv_s S \cdot H \cdot S$

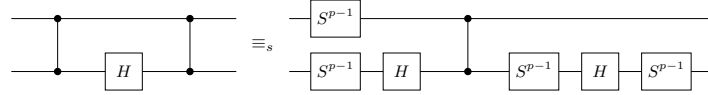
(Hadamard conjugation)

Conjugating S^k by H ; the instance used below is $H \cdot S^{-b/a} \cdot H$.



[selinger] $CZ \cdot H \cdot CZ \equiv_s S^{-1} \cdot H \cdot S^{-1} \cdot CZ \cdot H \cdot S^{-1} \cdot S^{-1\uparrow}$

(axiom selinger-c10/c11)



[Ex-slide] $g_{\downarrow} \cdot \text{Ex} \equiv_s \text{Ex} \cdot g_{\uparrow}$ (and $\uparrow \rightarrow \downarrow$), $CZ \cdot \text{Ex} = \text{Ex} \cdot CZ$

(swap lemmas)

A single-wire gate $g \in \{H, S, S^k, H^2\}$ slides across the swap Ex , changing wire; CZ (symmetric) passes unchanged. Exercised throughout the D/B chains.

[CZ-pow] $CZ \cdot CZ^k = CZ^{k+1}$, $CX \cdot CX^k = CX^{k+1}$

(generator order)

Combining adjacent same-control gates. Exercised in §3.7.

[X-CZ] $CZ \cdot X^{\uparrow} = X^{\uparrow} \cdot Z^{\downarrow} \cdot CZ$

(axiom rel-X $\uparrow$ -CZ)

How CZ conjugates a Pauli X ; underlies the two-qubit CZ relations.

[comm-CZ-S] $CZ \cdot S_{\downarrow} = S_{\downarrow} \cdot CZ$, $CZ \cdot S_{\uparrow} = S_{\uparrow} \cdot CZ$

(axioms comm-CZ-S $\downarrow/\uparrow$)

CZ commutes with a single-wire S . (Exercised in §3.5, §3.7.)

[HH-CZ] $CZ^{-a} \cdot H^2 \equiv_s H^2 \cdot CZ^a$

(lemma-semi-HH $\downarrow$ -CZ k)

$H^2 (= M[-1])$ inverts a CZ power. (Exercised in §3.4.)

[CZ-H-CZ] $CZ^{-a} \cdot H \cdot CZ \equiv_s H \cdot S_{\uparrow}^a \cdot S^{-1/a} \cdot CX^{-a} \cdot S^{1/a}$

(aux-CZ $^{-k}$ -H-CZ)

The two-qubit core (a CZ - H - CZ sandwich), the analogue of [selinger]. (Exercised in §3.7.)

[Euler] $H \cdot S^k \cdot H \equiv_s M[-1/k] \cdot S^{-k} \cdot H \cdot S^{-1/k}$

(lemma-Euler', the HS normal form)

The general Hadamard- S^k conjugation ([Hconj] is the $k = -1$ instance), producing an M box. (Exercised in $A \leftarrow H$ and $D \leftarrow H$, cases $a, b \neq 0$.)

[HH-CX] $CX^a \cdot H_{\uparrow}^2 \equiv_s H_{\uparrow}^2 \cdot CX^{-a}$

(up/down dual of [HH-CZ])

Used by $B \leftarrow H_{\uparrow}$. (Exercised in §3.8.)

[CX-S] $CX^b \cdot S_{\downarrow} \equiv_s S_{\downarrow} \cdot S_{\uparrow}^{b^2} \cdot CZ^{-b} \cdot CX^b$

(lemma-CX k -S)

Pushing $S_{\downarrow}$ through a CX power spawns a CZ — the two-qubit dual of [semi-MS]. (Exercised in §3.10.)

[M-CZ] $M[a] \cdot CZ \equiv_s CZ^{1/a} \cdot M[a]$

(axiom semi-M $\downarrow$ -CZ)

The metaplectic M box conjugates CZ to a CZ power (the exponent inverts in the Cir2Tikz $M[a]$ convention); the engine of the L and $A^{\uparrow} \leftarrow CZ$ relations.

[Ex-Ex-CZ] $\text{Ex} \cdot \text{Ex}^{\uparrow} \cdot CZ \equiv_s CZ^{\uparrow} \cdot \text{Ex} \cdot \text{Ex}^{\uparrow}$

(lemma-Ex-Ex $\uparrow$ -CZ)

The double swap carries a bottom-pair CZ to the top pair; the braiding at the heart of the three-qubit relations. (Exercised in §3.13.)

$$\mathbf{[CZ-CZ]} \quad CZ \cdot CZ^\uparrow \equiv_s CZ^\uparrow \cdot CZ, \quad CZ \cdot \text{Ex} = \text{Ex} \cdot CZ \quad (\text{axiom selinger-c12})$$

Disjoint CZ gates commute, and CZ slides through a same-pair Ex . (Exercised in §3.13.)

$$\mathbf{[CZ02-CZ-CZ]} \quad CZ_{02}^{-a} H CZ_{\uparrow}^{-c} H_{\uparrow} CZ \equiv_s H H_{\uparrow} CZ S^{-c/a} CX_{02}^{-a} S^{c/a} S_{\uparrow}^{-a/c} CX_{\uparrow}^{-c} S_{\uparrow}^{a/c} \quad (\text{lemma-CZ02}^k\text{-CZ}^l\text{-}\uparrow\text{-CZ})$$

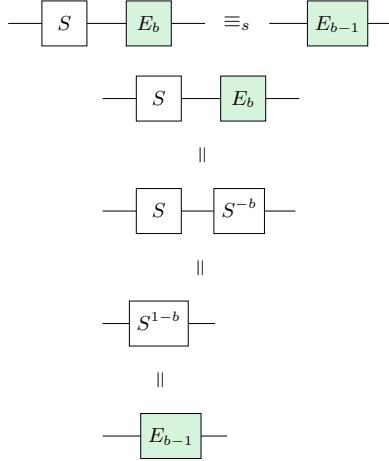
The three-wire core of $DD \leftarrow CZ$ (both heads nontrivial): the two-pair analogue of $\mathbf{[CZ-H-CZ]}$. (Exercised in §3.13.)

$$\mathbf{[A\uparrow-CZ]} \quad A^\uparrow[a, b] \cdot CZ \equiv_s (H CZ^{1/a} H^3) \cdot B[a, b] \cdot A^\uparrow[0, -a] \quad (\text{lemma-A-CZ-2})$$

How CZ passes a top-wire A box: it descends, spawning a B box. The composite sub-relation behind $L \leftarrow CZ$ cases with an $A^\uparrow$ factor (itself built from $\mathbf{[M-CZ]}$, $\mathbf{[Euler]}$). Proven in `N/BR/Two/A-CZ.agda`.

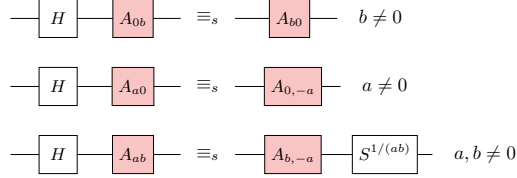
3 Layer 1 — Box relations

$$\mathbf{3.1} \quad E \leftarrow S: \quad S \cdot E[b] \equiv_s E[b-1]$$

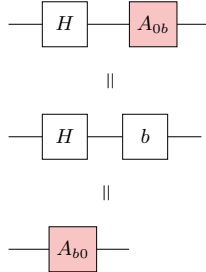


Trace. $S \cdot E[b] = S \cdot S^{-b}$ (expand $E[b] = S^{-b}$) $= S^{1-b}$ (combine powers) $= E[b-1]$ (fold). Purely definitional.

$$\mathbf{3.2} \quad A \leftarrow H: \quad H \cdot A[\mathbf{v}] \equiv_s \text{dir} \cdot A[\mathbf{v}']$$

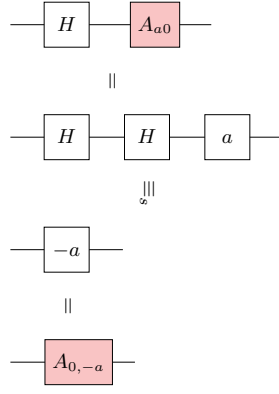


Case $a = 0, b \neq 0$ ($H \cdot A[0, b] = A[b, 0]$).



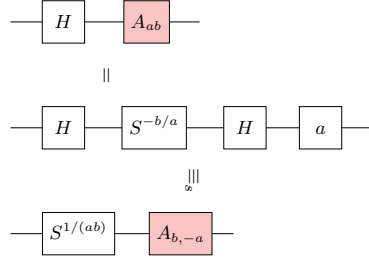
Trace. expand $A[0, b] = M[b] = ;$ the word $[H, M[b]]$ folds to $A[b, 0]$. Definitional.

Case $a \neq 0, b = 0$ ($H \cdot A[a, 0] \equiv_s A[0, -a]$).



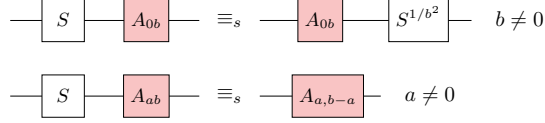
Trace. expand $A[a, 0] = M[a] \cdot H =$ gives $[H, H, M[a]]$; then $\equiv_s$ by [HHM] (so $H^2 M[a] = M[-a]$), fold $A[0, -a]$.

Case $a, b \neq 0$ ($H \cdot A[a, b] \equiv_s S^{1/(ab)} \cdot A[b, -a]$).

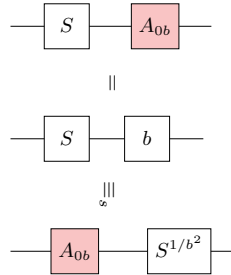


Trace. expand $A[a, b] = [H, S^{-b/a}, H, M[a]]$; then $\equiv_s$ by [Hconj] on the $H S^{-b/a} H$ block, and [M-mul] to recombine; fold $S^{1/(ab)} \cdot A[b, -a]$.

3.3 $A \leftarrow S$: $S \cdot A[\mathbf{v}] \equiv_s \text{dir} \cdot A[\mathbf{v}']$

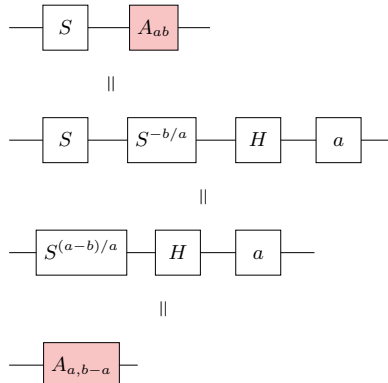


Case $a = 0$ ($S \cdot A[0, b] \equiv_s S^{1/b^2} \cdot A[0, b]$).



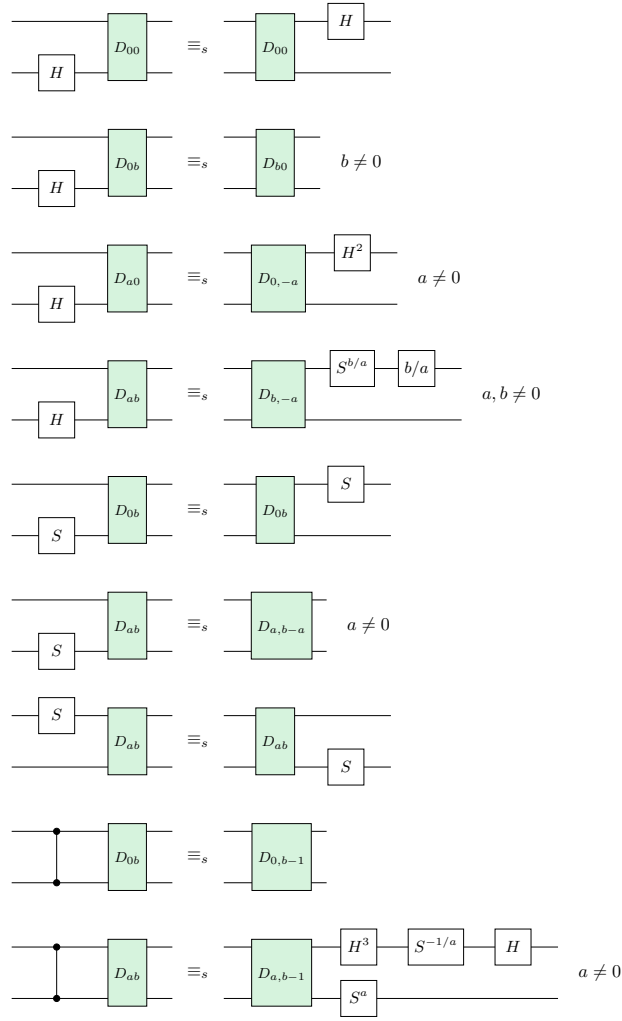
Trace. expand $A[0, b] = M[b]$; then $\equiv_s$ directly by [semi-MS]; fold.

Case $a \neq 0$ ($S \cdot A[a, b] = A[a, b - a]$).

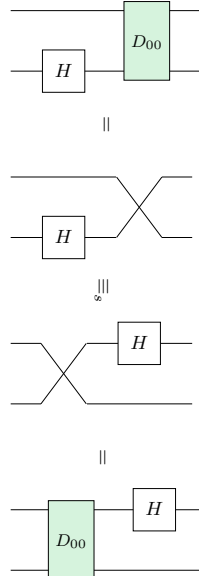


Trace. expand, combine $S \cdot S^{-b/a} = S^{(a-b)/a}$, fold. Definitional.

3.4 $D \leftarrow H$: $H_{\downarrow} \cdot D[\mathbf{v}] \equiv_s \text{dir} \cdot D[\mathbf{v}']$

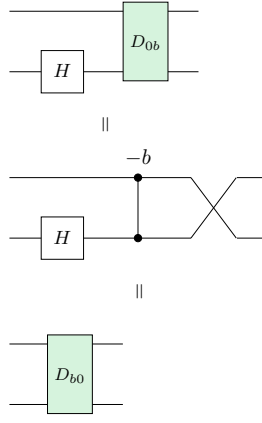


Case $a = b = 0$ ($H_{\downarrow} \cdot D[0, 0] \equiv_s D[0, 0] \cdot H_{\uparrow}$).



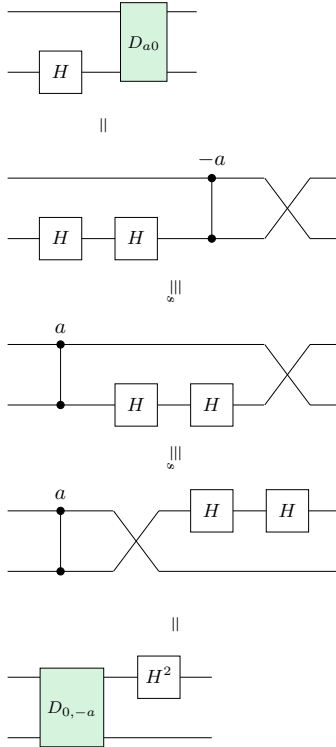
Trace. expand $D[0, 0] = \text{Ex}$; the middle step $H_{\downarrow} \cdot \text{Ex} \equiv_s \text{Ex} \cdot H_{\uparrow}$ is [Ex-slide]; fold.

Case $a = 0, b \neq 0$ ($H_{\downarrow} \cdot D[0, b] = D[b, 0]$).



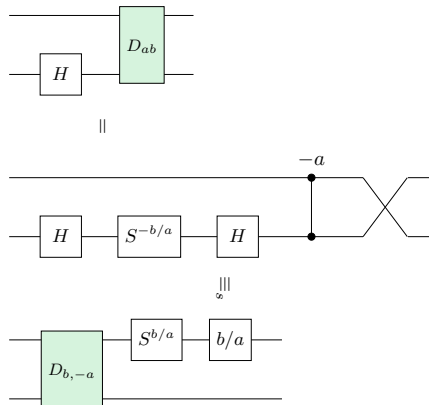
Trace. expand $D[0, b] = [CZ^{-b}, \text{Ex}]$; $[H, CZ^{-b}, \text{Ex}]$ folds to $D[b, 0]$. Definitional.

Case $a \neq 0, b = 0$ ($H_{\downarrow} \cdot D[a, 0] \equiv_s H_{\uparrow}^2 \cdot D[0, -a]$).



Trace. expand $D[a, 0]$; [HH-CZ] ($CZ^{-a}H^2 = H^2CZ^a$); [Ex-slide] (H^2 across Ex); fold $D[0, -a]$.

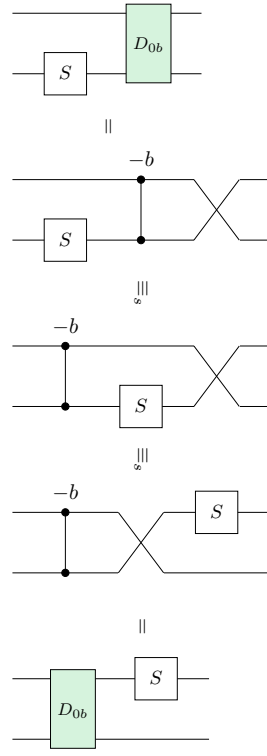
Case $a, b \neq 0$ ($H_{\downarrow} \cdot D[a, b] \equiv_s M_{\uparrow}[b/a] \cdot S_{\uparrow}^{b/a} \cdot D[b, -a]$).



Trace. expand $D[a, b]$; [Euler] on the $H S^{-b/a} H$ block (creates the M), then [comm-CZ-S]/[Ex-slide] cleanup and [M-mul]; fold.

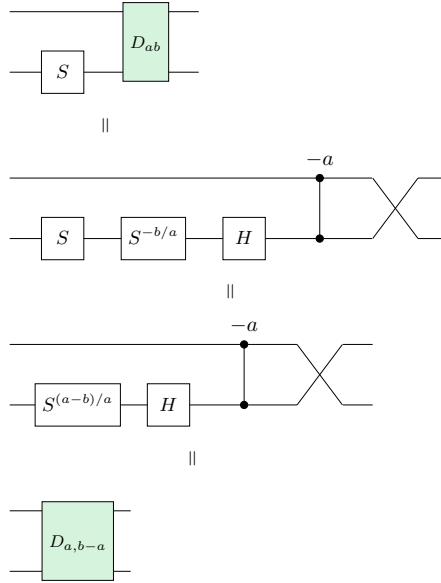
3.5 $D \leftarrow S_{\downarrow}$: $S_{\downarrow} \cdot D[\mathbf{v}] \equiv_s \text{dir} \cdot D[\mathbf{v}']$

Case $a = 0$ ($S_{\downarrow} \cdot D[0, b] \equiv_s D[0, b] \cdot S_{\uparrow}$).



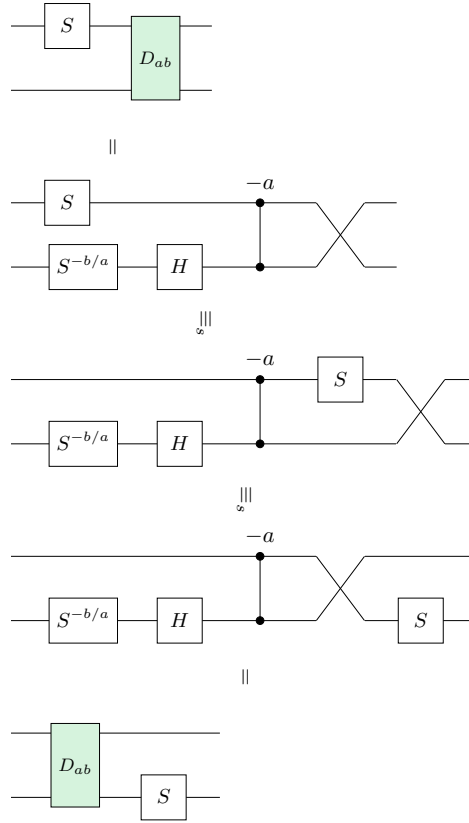
Trace. expand $D[0, b]$; [comm-CZ-S] (S past CZ); [Ex-slide] ($S_{\downarrow}$ across Ex $\rightarrow S_{\uparrow}$); fold.

Case $a \neq 0$ ($S_{\downarrow} \cdot D[a, b] = D[a, b - a]$).



Trace. expand, combine $S \cdot S^{-b/a} = S^{(a-b)/a}$, fold. Definitional.

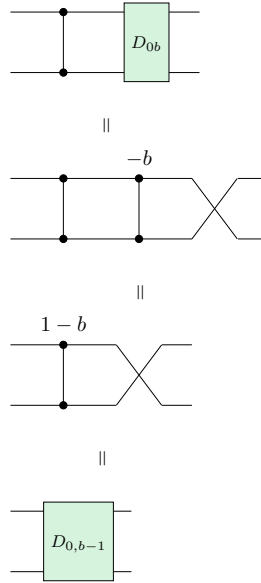
3.6 $D \leftarrow S_{\uparrow}$: $S_{\uparrow} \cdot D[\mathbf{v}] \equiv_s D[\mathbf{v}] \cdot S_{\downarrow}$



Trace. expand $D[a, b]$; $S_{\uparrow}$ commutes left through the bottom-wire gates ([comm-CZ-S] for the CZ); [Ex-slide] ($S_{\uparrow}$ across $\text{Ex} \rightarrow S_{\downarrow}$); fold.

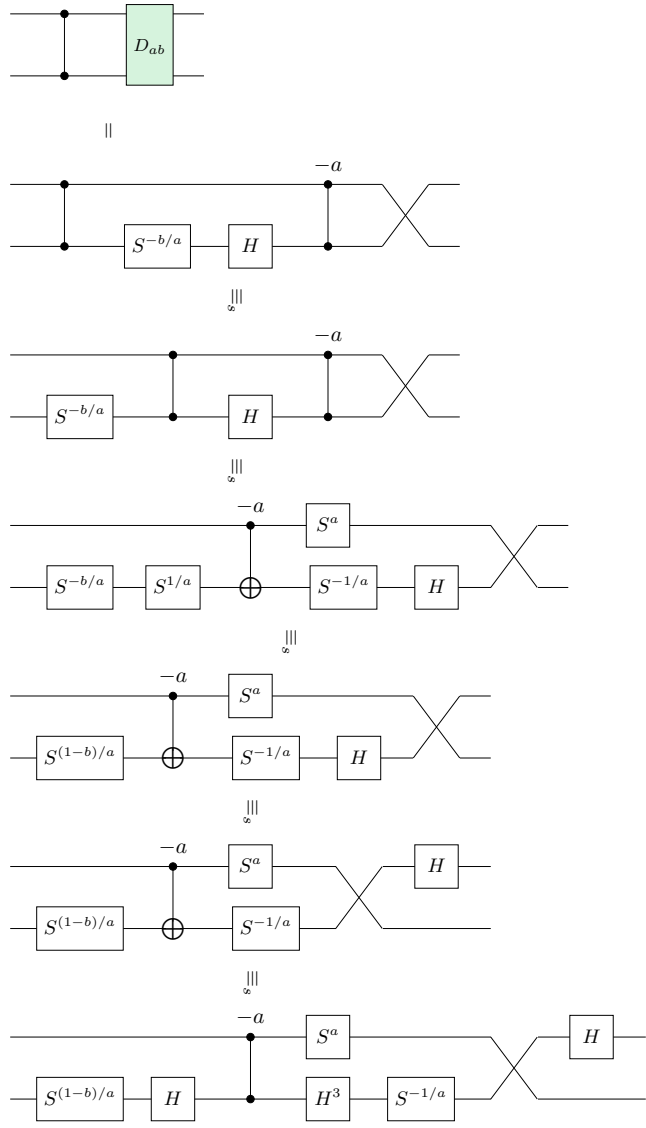
3.7 $D \leftarrow CZ$: $CZ \cdot D[\mathbf{v}] \equiv_s \text{dir} \cdot D[\mathbf{v}']$

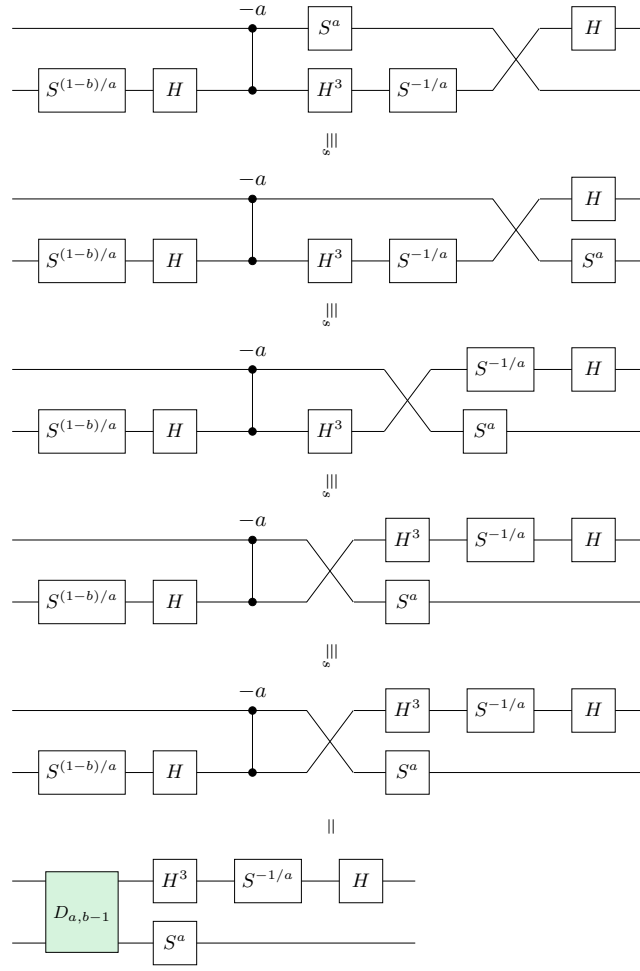
Case $a = 0$ ($CZ \cdot D[0, b] = D[0, b-1]$).



Trace. expand; $CZ \cdot CZ^{-b} = CZ^{1-b}$ by [CZ-pow]; fold. (Definitional after [CZ-pow].)

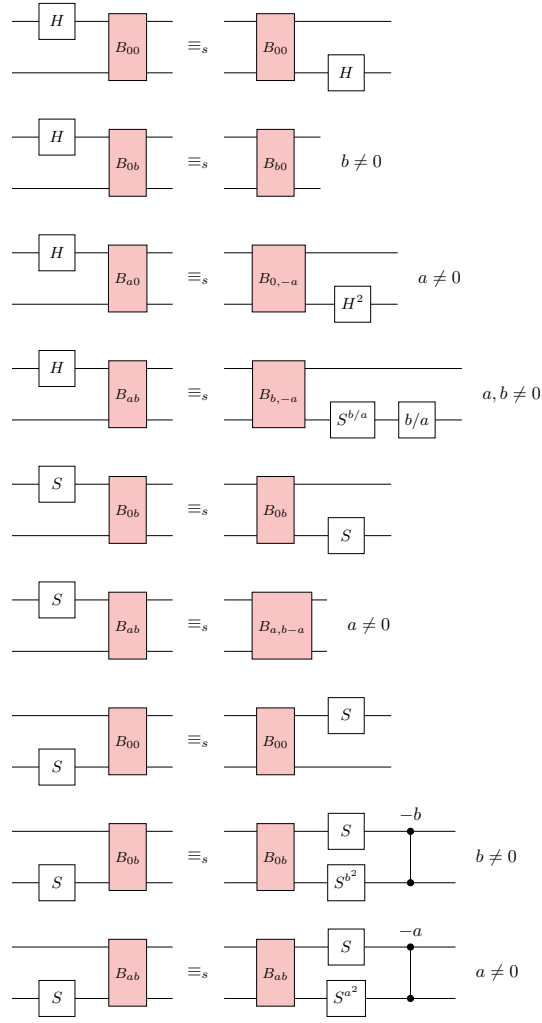
Case $a \neq 0$ ($CZ \cdot D[a, b] \equiv_s S_{\downarrow}^a \cdot H_{\uparrow} \cdot S_{\uparrow}^{-1/a} \cdot H_{\uparrow}^3 \cdot D[a, b-1]$).





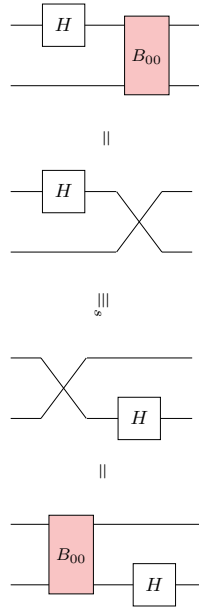
Trace (fully expanded to axioms). expand $D[a, b] = [S^{-b/a}, H, CZ^{-a}, \text{Ex}]$; [comm-CZ-S] moves CZ left past $S^{-b/a}$; [CZ-H-CZ] rewrites the core $CZ^{-a} H CZ = H S_{\uparrow}^a S^{-1/a} C X^{-a} S^{1/a}$ (spawning the $C X^{-a}$); [CZ-pow] merges $S^{1/a} S^{-b/a} = S^{(1-b)/a}$; [Ex-slide] floats H up ($\text{Ex } H = H_{\uparrow} \text{Ex}$); $C X^{-a} = H^3 C Z^{-a} H$ rebuilds a CZ ([CZ-H-CZ] read backwards); three more [Ex-slide]s carry $S_{\uparrow}^a$, $S_{\downarrow}^{-1/a}$ and $H_{\downarrow}^3$ across Ex ; [comm-CZ-S] commutes $S_{\downarrow}^a$ past $H_{\uparrow}$; fold $\text{dir}^{\uparrow} = H_{\uparrow} S_{\uparrow}^{-1/a} H_{\uparrow}^3$ and the box $D[a, b-1]$, leaving $S_{\downarrow}^a$.

3.8 $B \leftarrow H_{\uparrow}$: $H_{\uparrow} \cdot B[\mathbf{v}] \equiv_s \text{dir} \cdot B[\mathbf{v}']$



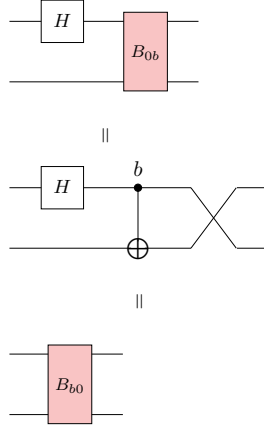
These mirror $D \leftarrow H$ under the up/down swap.

Case $a = b = 0$ ($H_{\uparrow} \cdot B[0, 0] \equiv_s B[0, 0] \cdot H_{\downarrow}$).



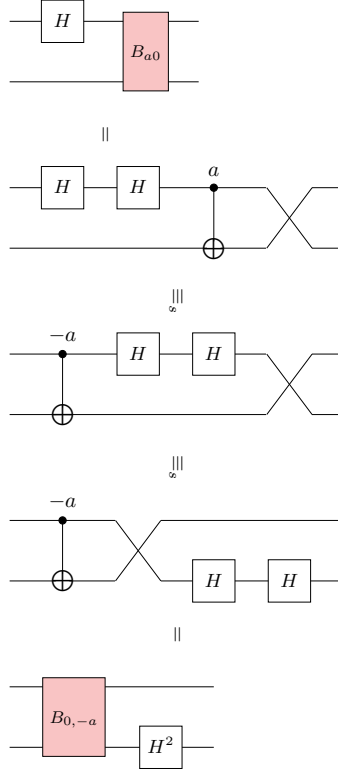
Trace. expand $B[0, 0] = \text{Ex}$; middle step is [Ex-slide]; fold.

Case $a = 0, b \neq 0$ ($H_{\uparrow} \cdot B[0, b] = B[b, 0]$).



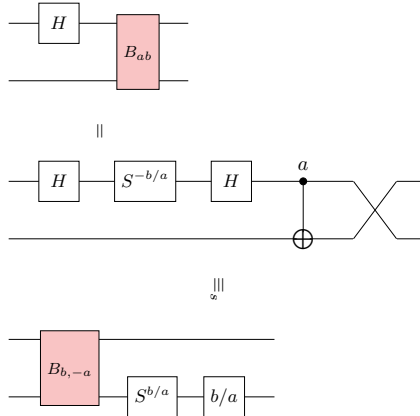
Trace. expand $B[0, b] = [CX^b, \text{Ex}]$; folds to $B[b, 0]$. Definitional.

Case $a \neq 0, b = 0$ ($H_{\uparrow} \cdot B[a, 0] \equiv_s H_{\downarrow}^2 \cdot B[0, -a]$).



Trace. expand $B[a, 0]$; [HH-CX] ($CX^a H_{\uparrow}^2 = H_{\uparrow}^2 CX^{-a}$); [Ex-slide] ($H_{\uparrow}^2$ across Ex $\rightarrow H_{\downarrow}^2$); fold $B[0, -a]$.

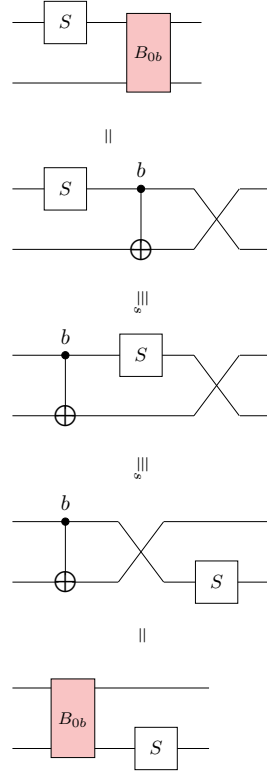
Case $a, b \neq 0$ ($H_{\uparrow} \cdot B[a, b] \equiv_s M_{\downarrow}[b/a] \cdot S_{\downarrow}^{b/a} \cdot B[b, -a]$).



Trace. expand $B[a, b]$; [Euler] on the $H S^{-b/a} H$ block, then [Ex-slide] and [M-mul]; fold. (Up/down dual of $D \leftarrow H$, $a, b \neq 0$.)

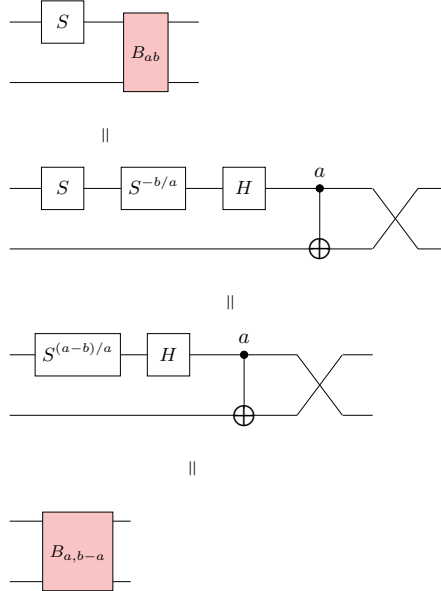
3.9 $B \leftarrow S_{\uparrow}:$ $S_{\uparrow} \cdot B[\mathbf{v}] \equiv_s \text{dir} \cdot B[\mathbf{v}']$

Case $a = 0$ ($S_{\uparrow} \cdot B[0, b] \equiv_s B[0, b] \cdot S_{\downarrow}$).



Trace. expand $B[0, b]$; $S_{\uparrow}$ commutes past CX (control wire); [Ex-slide] ($S_{\uparrow}$ across $\text{Ex} \rightarrow S_{\downarrow}$); fold. (Dual of $D \leftarrow S_{\downarrow}$, $a = 0$.)

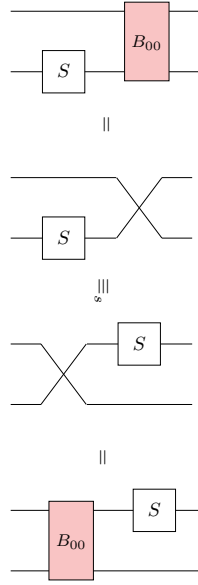
Case $a \neq 0$ ($S_{\uparrow} \cdot B[a, b] = B[a, b - a]$).



Trace. expand, combine S -powers, fold. Definitional.

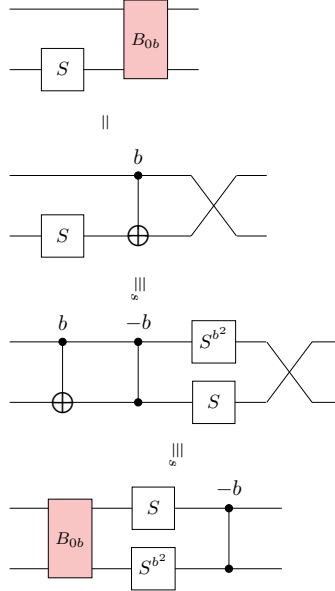
3.10 $B \leftarrow S_{\downarrow}:$ $S_{\downarrow} \cdot B[\mathbf{v}] \equiv_s \text{dir} \cdot B[\mathbf{v}']$

Case $a = b = 0$ ($S_{\downarrow} \cdot B[0, 0] \equiv_s B[0, 0] \cdot S_{\uparrow}$).



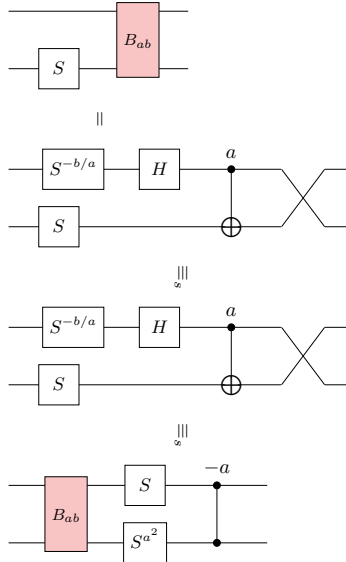
Trace. expand $B[0, 0] = \text{Ex}$; middle step $S_\downarrow \cdot \text{Ex} \equiv_s \text{Ex} \cdot S_\uparrow$ is [Ex-slide]; fold.

Case $a = 0, b \neq 0$ ($S_\downarrow \cdot B[0, b] \equiv_s CZ^{-b} \cdot S_\downarrow^{b^2} \cdot S_\uparrow \cdot B[0, b]$).



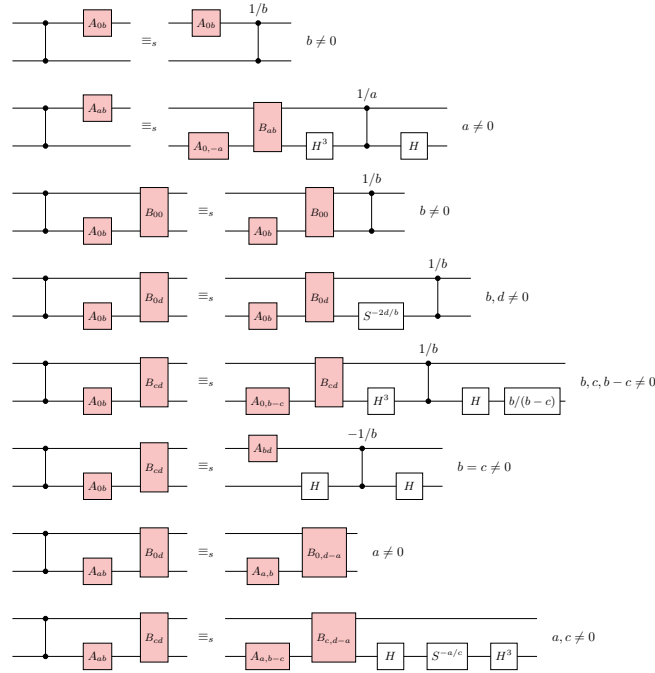
Trace. expand $B[0, b] = [CX^b, \text{Ex}]$; [CX-S] ($CX^b S_\downarrow = S_\downarrow S_\uparrow^{b^2} CZ^{-b} CX^b$, spawning the CZ^{-b}); [Ex-slide] moves $S_\downarrow, S_\uparrow^{b^2}, CZ^{-b}$ past Ex; fold.

Case $a \neq 0$ ($S_\downarrow \cdot B[a, b] \equiv_s CZ^{-a} \cdot S_\downarrow^{a^2} \cdot S_\uparrow \cdot B[a, b]$).



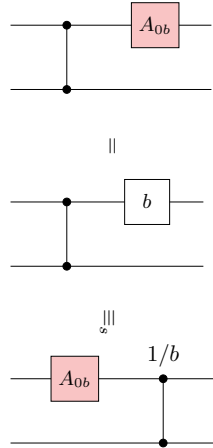
Trace. expand $B[a, b]$; $S_\downarrow$ commutes past the $(H S^{-b/a})_\uparrow$ head, reducing to the $a = 0$ case: [CX-S] then [Ex-slide]; fold.

3.11 $L \leftarrow CZ$: $CZ \cdot L[\mathbf{v}] \equiv_s \text{dir} \cdot L[\mathbf{v}']$



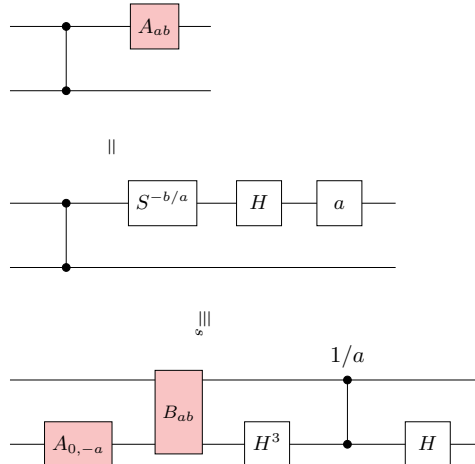
An L box is $A^\dagger$ (top wire only) or $A \cdot B$; the eight cases split on which factors are present and whether their first indices vanish or coincide. In every case the chain shows the box-expansion, then routes CZ to the right, then folds.

Case 1 $(CZ \cdot A^\dagger[0, b], b \neq 0)$.



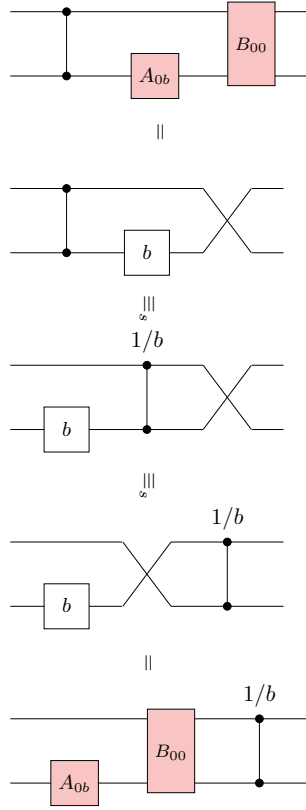
Trace (to axioms). $A^\dagger[0, b] = M^\dagger[b]$; the single $\equiv_s$ is $[M-CZ]$ on the top wire ($M^\dagger[b] CZ = CZ^{1/b} M^\dagger[b]$); fold.

Case 2 $(CZ \cdot A^\dagger[a, b], a \neq 0)$.



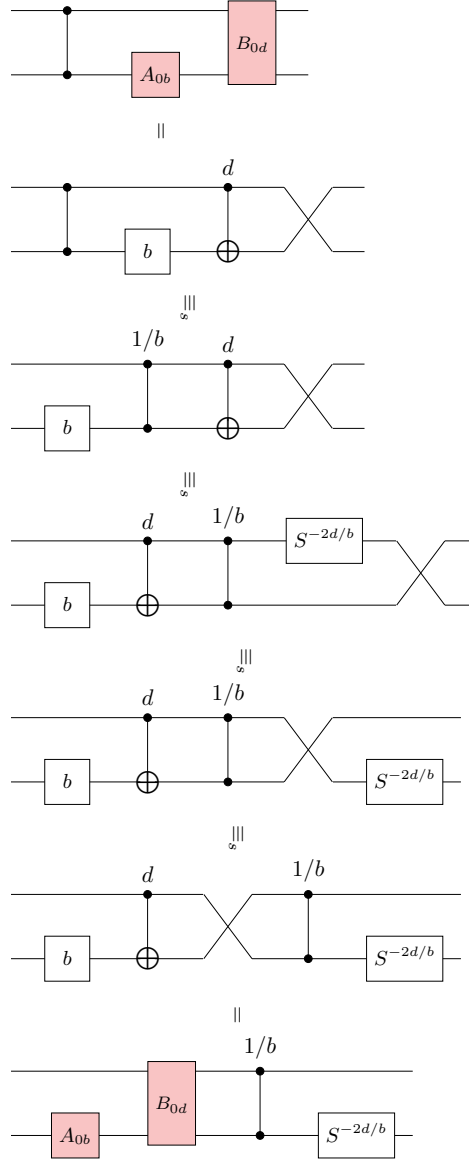
Trace (to named lemmas). expand $A^\dagger[a, b]$; the single $\equiv_s$ is $[A^\dagger-CZ]$ ($\text{dir} = H CZ^{1/a} H^3$, spawning $B[a, b]$); fold. ($[A^\dagger-CZ]$ unfolds to $[M-CZ] + [\text{Euler}]$.)

Case 3 $(CZ \cdot A[0, b] \cdot B[0, 0])$.



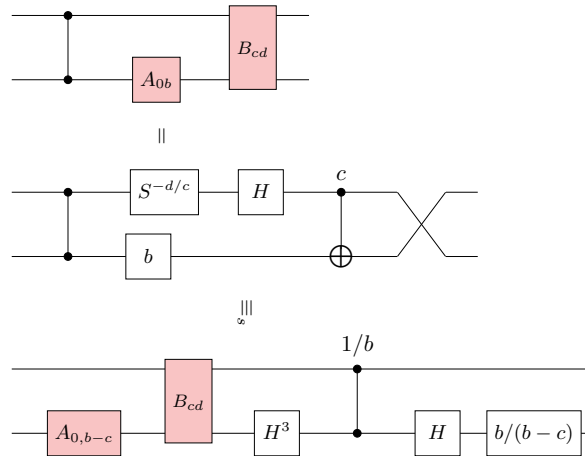
Trace (fully expanded to axioms). $B[0, 0] = \text{Ex}$, $A[0, b] = M[b]$; [M-CZ] ($M[b] CZ = CZ^{1/b} M[b]$); [Ex-slide] (CZ past Ex); fold.

Case 4 $(CZ \cdot A[0, b] \cdot B[0, d], b, d \neq 0)$.



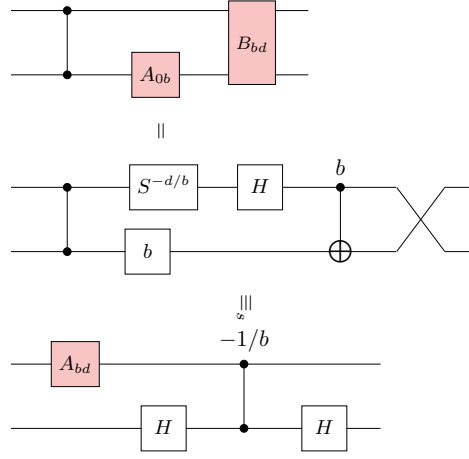
Trace (fully expanded to axioms). expand $A[0, b] = M[b]$, $B[0, d] = [CX^d, \text{Ex}]$; [M-CZ] ($M[b] CZ = CZ^{1/b} M[b]$); [X-CZ] (CX^d through CZ spawns $S_{\uparrow}^{-2d/b}$); [Ex-slide] ($S_{\uparrow} \rightarrow S_{\downarrow}$ and CZ across Ex); fold.

Case 5 ($CZ \cdot A[0, b] \cdot B[c, d]$, $b \neq c$).



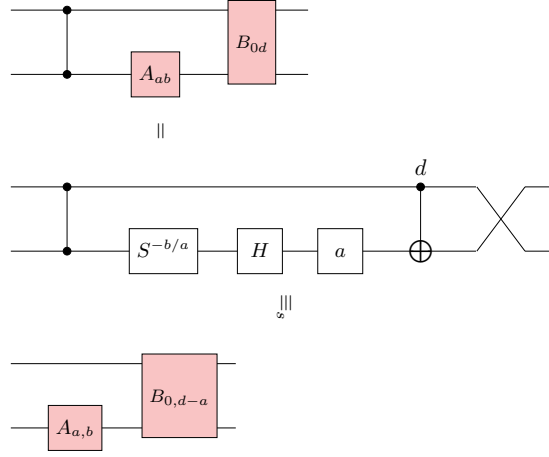
Trace (to named lemmas; L-CZ.agda:588). expand both factors ($B[c, d]$ via $CX'^c = H^3 CZ^c H$); [M-CZ] and the top-wire $M^{\uparrow}\text{-CZ}$ pull CZ in; [semi-MS]/[Hconj] normalise the A head; the $b \neq c$ generic core fires lemma- $\sqsupset\text{-CZ}^y$ (a CZ -conjugation producing $M CZ \sqsupset M$); [HH-CZ] and [Ex-slide] clean up; fold. (Full circuit inlining deferred: 98-step proof with the auxiliary $\sqsupset$ gate.)

Case 6 ($CZ \cdot A[0, b] \cdot B[b, d]$, $b = c$ coincidence).



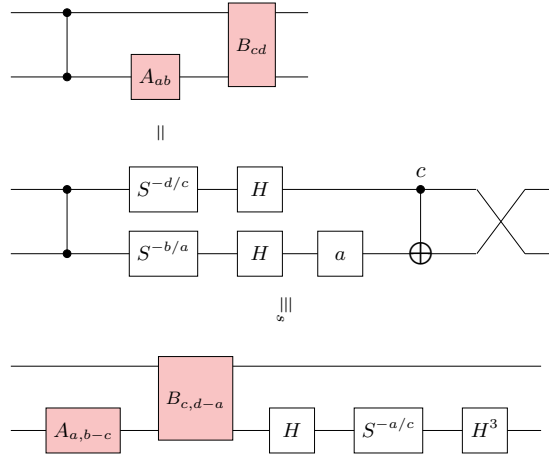
Trace (to named lemmas; L-CZ.agda:543). the $b = c$ coincidence: conjugating by $(H CZ^{1/b} H^3)(\cdot) CZ^{-1}$ and using [A $\uparrow$ -CZ] backwards collapses the $A \cdot B$ pair to a single $A^\uparrow[b, d]$; dir = $(H CZ^{1/b} H^3)^{-1} H^2$. (Reduction proof; circuit inlining deferred.)

Case 7 ($CZ \cdot A[a, b] \cdot B[0, d]$, $a \neq 0$).



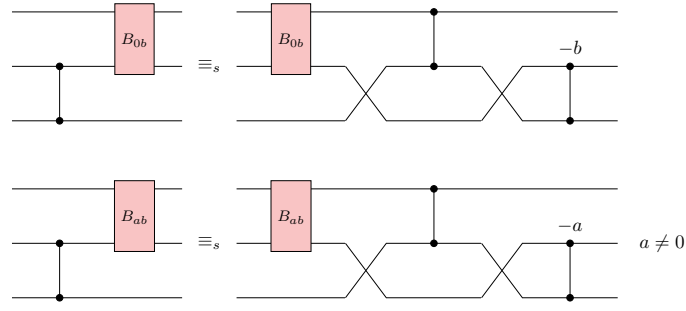
Trace (to named lemmas; L-CZ.agda:414, dir = ε). expand $A[a, b] = M[a] H S^{-b/a}$, $B[0, d]$; [comm-CZ-S] then [M-CZ] ($M[a] CZ = CZ^{1/a} M[a]$); [CZ-pow] merges the two CZ powers; [HH-CZ] and [semi-MS] reabsorb everything into the boxes — CZ is fully absorbed (dir = ε), giving $A[a, b] \cdot B[0, d - a]$. (Circuit inlining deferred: uses $CX' = H^3 CZ H$.)

Case 8 ($CZ \cdot A[a, b] \cdot B[c, d]$, $a, c \neq 0$).

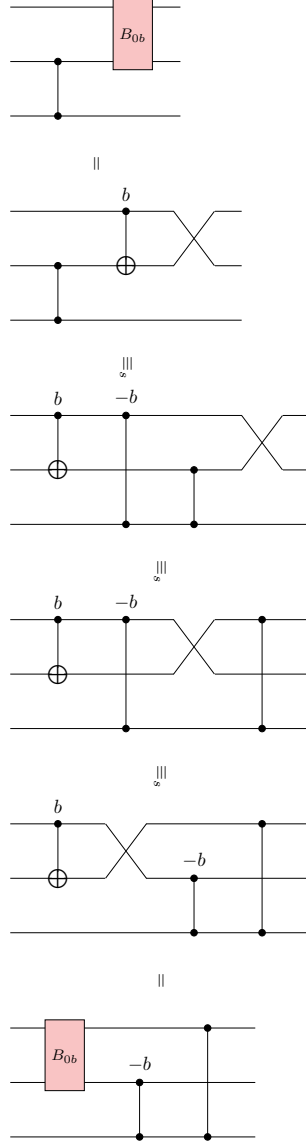


Trace (to named lemmas; L-CZ.agda:278). the general case, ~ 35 steps: [comm-CZ-S], then [CZ-H-CZ] (aux-CZ $^{-k}$ -H-CZ) on the A - B junction, [Euler] (lemma-Euler') on both heads, [HH-CZ], [semi-MS] and many [Ex-slide]s; dir = $H^3 S^{-a/c} H$. (Full circuit inlining deferred.)

3.12 $B^\uparrow \leftarrow CZ$: $CZ \cdot B^\uparrow[\mathbf{v}] \equiv_s \text{dir} \cdot B^\uparrow[\mathbf{v}]$



Case $a = 0$.

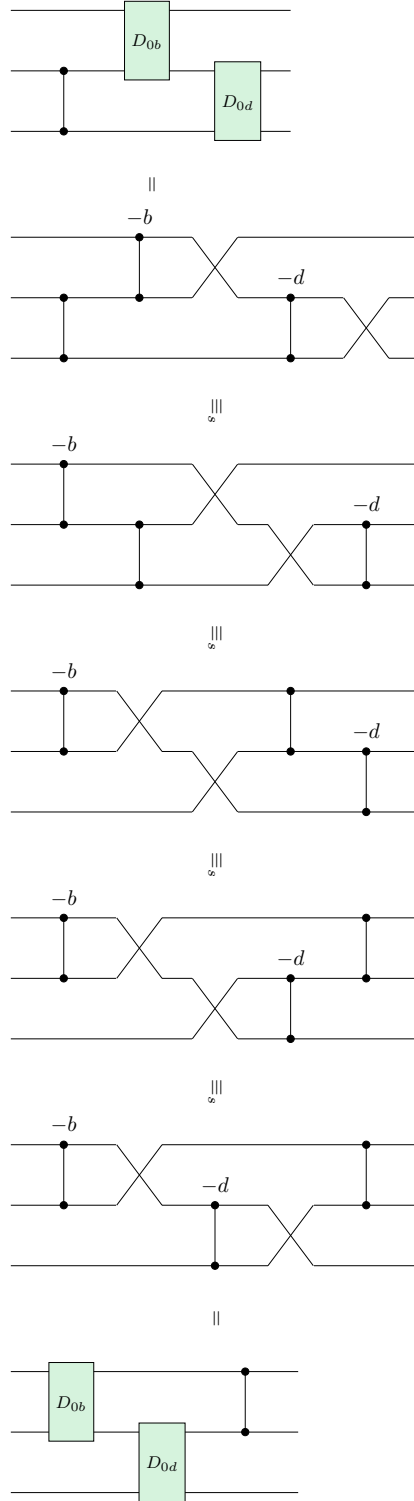


Trace (to axioms). expand $B^\uparrow[0, b] = [CX^b, \text{Ex}]^\uparrow$; [X-CZ] pushes $CX^\uparrow$ past CZ , spawning the long-range CZ_{02}^{-b} ; [Ex-slide] sends $\text{Ex}^\uparrow \cdot CZ = CZ_{02} \cdot \text{Ex}^\uparrow$, then $\text{Ex}^\uparrow \cdot CZ_{02}^{-b} = CZ^{-b} \cdot \text{Ex}^\uparrow$; fold $\text{dir} = CZ_{02} \cdot CZ^{-b}$.

Case $a \neq 0$.

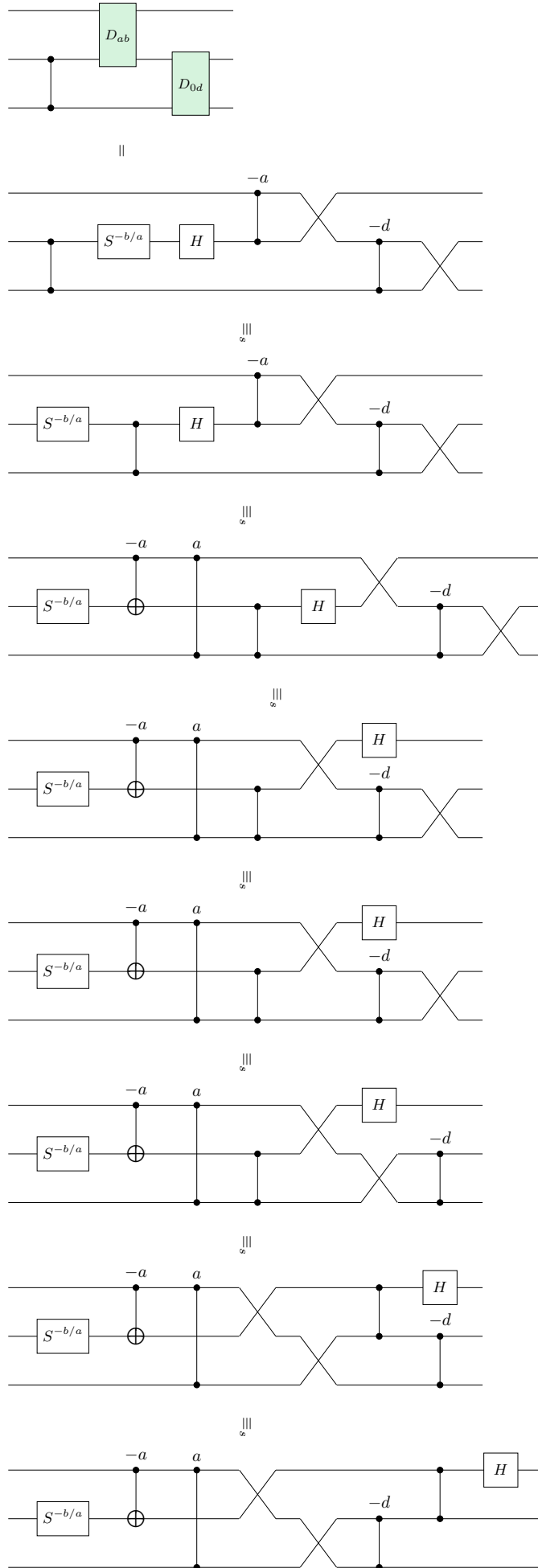
The four cases split on $a_1, a_2 = 0$ or $\neq 0$; each routes CZ through both D boxes by the braiding [Ex-Ex-CZ], with the $a_i \neq 0$ heads contributing the $D \leftarrow CZ$ dir (§3.7, §3.7).

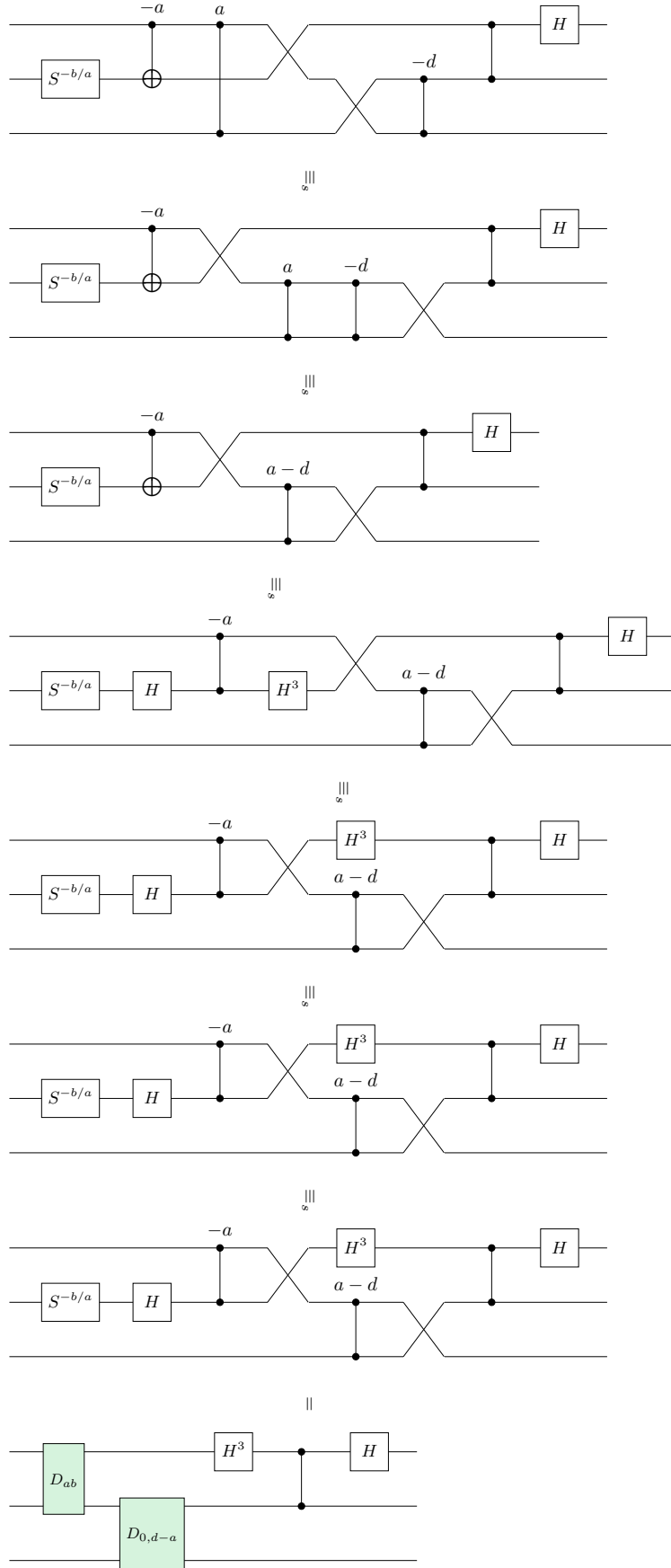
Case $a_1 = a_2 = 0$ (fully expanded to axioms).



Trace. expand both $D[0, \cdot] = [CZ^\cdot, \text{Ex}]$; [CZ-CZ] slides the outer CZ inward past $CZ_{\uparrow}^{-b}$ and Ex; [Ex-Ex-CZ] braids it to the top pair as $CZ^\dagger$; [CZ-CZ] (`selinger-c12`) commutes it past CZ^{-d} ; [CZ-CZ] slides it out; fold $CZ^\dagger \cdot DD[\mathbf{v}, \mathbf{v}]$.

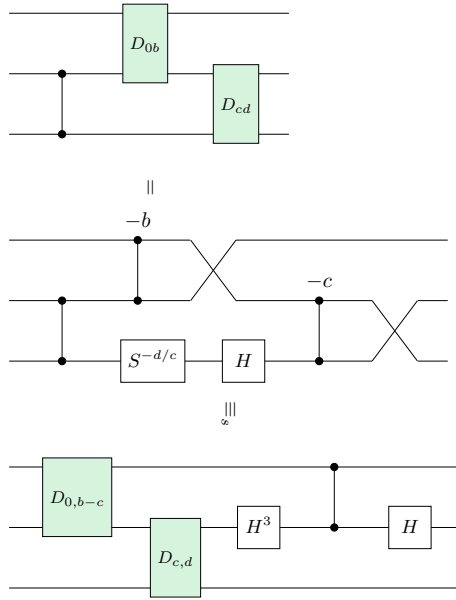
Case $a_1 = 0, a_2 \neq 0$ (upper box nontrivial; fully expanded to axioms).





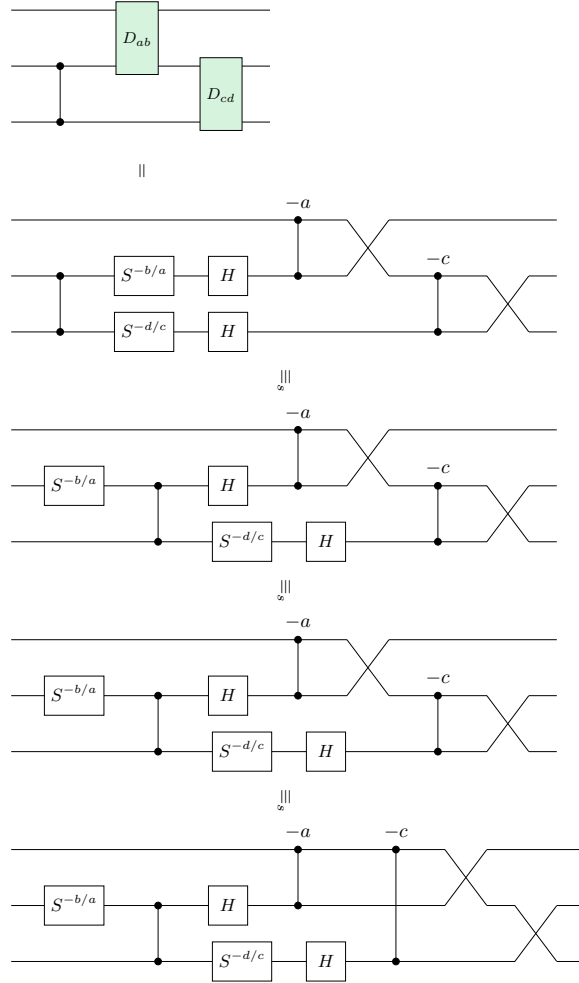
Trace. expand both boxes; [comm-CZ-S] pulls the upper $S^\uparrow$ past CZ ; the cross identity $CZ_\uparrow^{-a} H_\uparrow CZ = H_\uparrow CZ CZ_{02}^a CX_\uparrow^{-a}$ ([CZ-H-CZ] on the 0-2 pair) spawns the long-range CZ_{02} ; [Ex-slide] and [comm-Ex-CZ] float the upper H /box out of the way; [Ex-Ex-CZ] braids CZ to the top, [CZ-CZ] commutes it through, CZ_{02} collapses to CZ across $\text{Ex}^\uparrow$ and merges by [CZ-pow]; finally $CX_\uparrow^{-a} = H_\uparrow^3 CZ_\uparrow^{-a} H_\uparrow$ rebuilds the upper D box and the $H_\uparrow^{(3)}$ conjugation leaves $\text{dir} = H^\uparrow CZ H^{3\uparrow}$; fold.

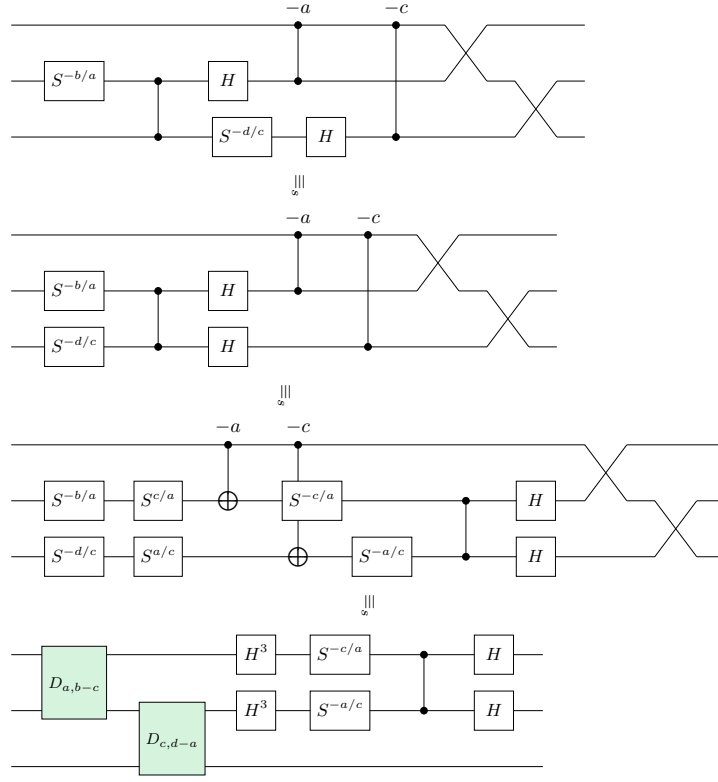
Case $a_1 \neq 0, a_2 = 0$ (lower box nontrivial; reduction to the previous case).



Trace. this is the mirror of §3.13: conjugating the whole equation by $\text{Ex}^\uparrow(\cdot)\text{Ex}$ swaps the two D boxes ([Ex-Ex-CZ] / Yang-Baxter, lemma-swap-DD), turning it into the $a_1 = 0, a_2 \neq 0$ case already expanded above; conjugating back gives the result with $\text{dir} = H_\downarrow CZ H_\downarrow^3$.

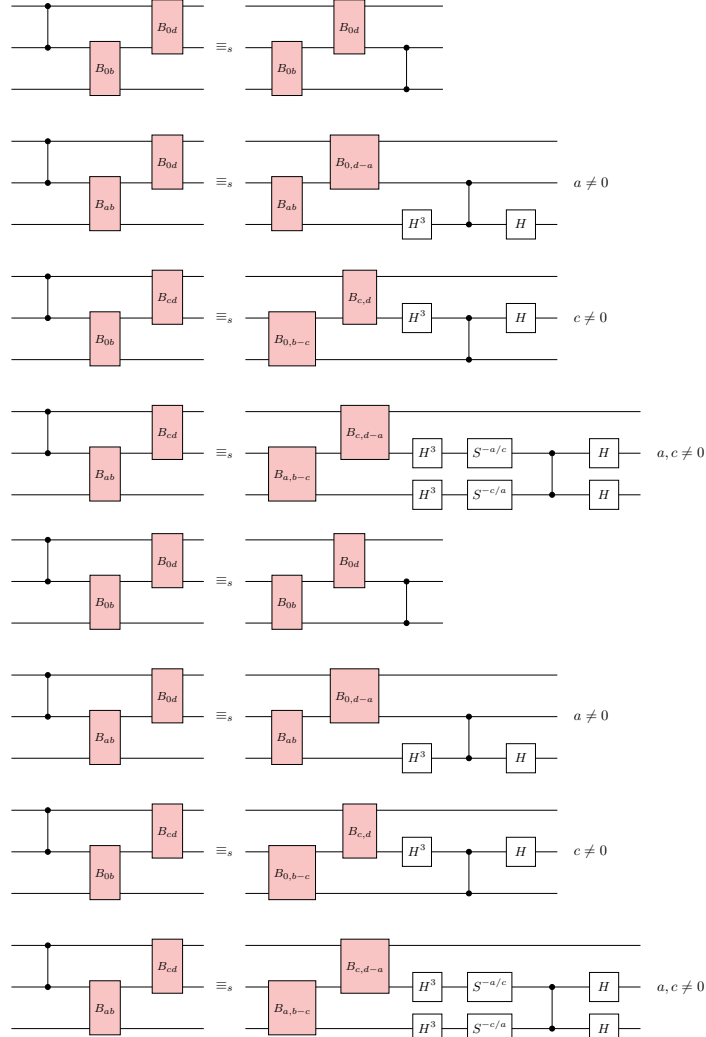
Case $a_1, a_2 \neq 0$ (both heads nontrivial; fully expanded to axioms).





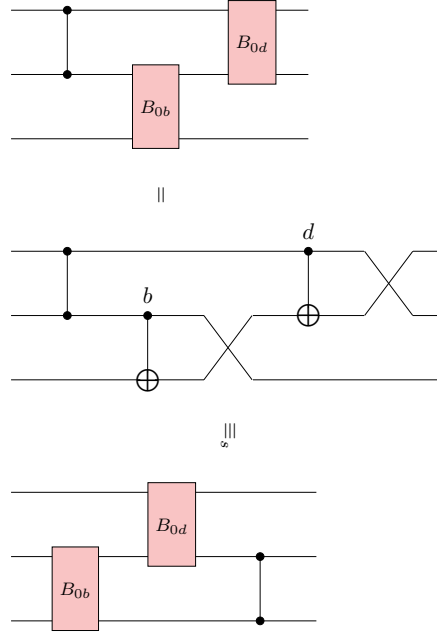
Trace. expand both D boxes; [comm-CZ-S] pulls each S head clear of CZ ; [Ex-slide]/[comm-Ex-CZ] bring the two CZ powers and the swaps together (one becomes the long-range CZ_{02}); then the three-wire core [CZ02-CZ-CZ] fires, producing the CX_{02} , $CX_{\uparrow}$ and the four S rescalings; a final mechanical pass of [Ex-slide]_s, [CZ-pow] S -merges and $CX = H^3 CZ H$ folds rebuilds $DD[\mathbf{v}']$ and leaves $\text{dir}^{\uparrow} = H_{\uparrow} H_{\uparrow\uparrow} CZ_{\uparrow} S_{\uparrow}^{-a/c} H_{\uparrow}^3 S_{\uparrow\uparrow}^{-c/a} H_{\uparrow\uparrow}^3$.

3.14 $BB \leftarrow CZ^{\uparrow}: \quad CZ^{\uparrow} \cdot BB[\mathbf{v}_1, \mathbf{v}_2] \equiv_s \text{dir}^{\downarrow} \cdot BB[\mathbf{v}'_1, \mathbf{v}'_2]$



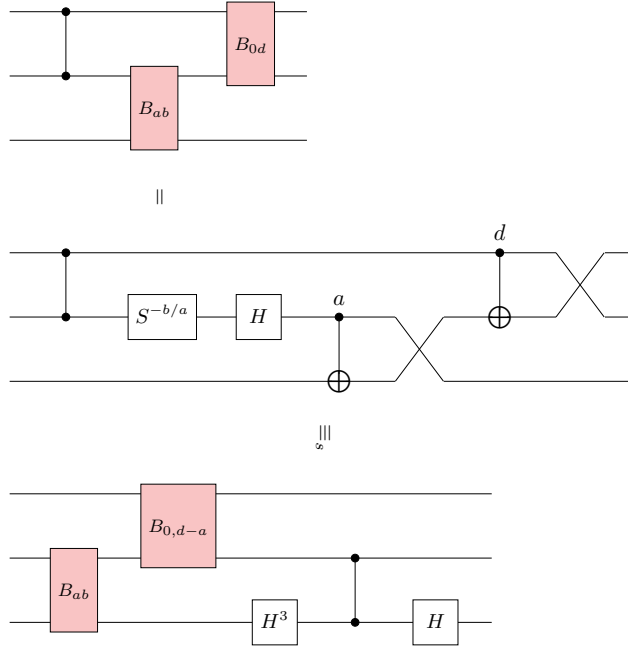
The up/down dual of $DD \leftarrow CZ$: the four cases split on the first index of the lower box B_1 (a_1) and the upper box B_2 (a_2).

Case $a_1 = 0, a_2 = 0$.



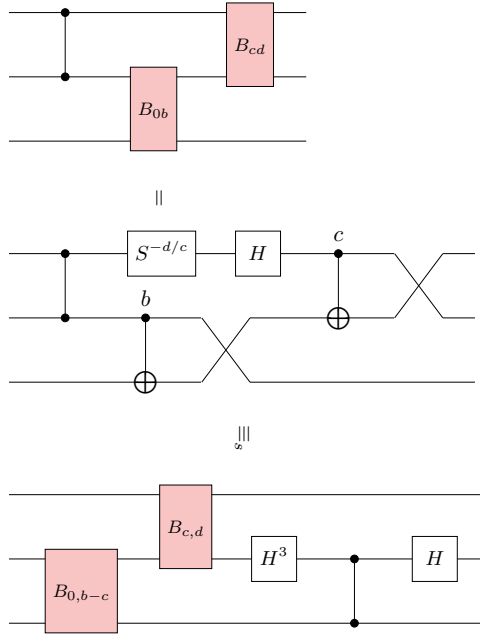
Trace. $BB \leftarrow CZ^\uparrow$ is the exact up/down dual of $DD \leftarrow CZ$: $\text{dir} = \text{dual}(\text{dir}_{DD}^{\text{rev}})$, and the proof conjugates the corresponding DD chain (§3.13) by the H^3/Ex duality bridge ($BB \sim DD$). So every step is the dual of the DD case 1 derivation.

Case $a_1 \neq 0, a_2 = 0$ (lower box nontrivial).



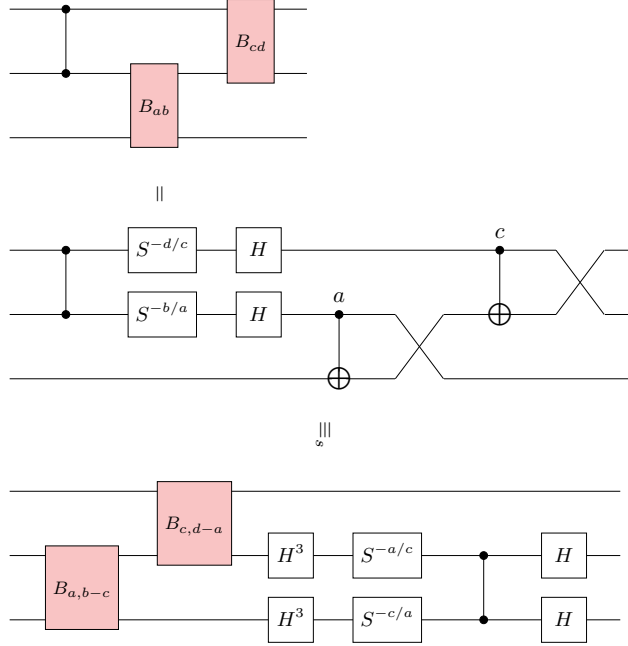
Trace. dual of DD case §3.13 under the up/down bridge ($BB \sim DD$): conjugate that derivation by H^3/Ex .

Case $a_1 = 0, a_2 \neq 0$ (upper box nontrivial).



Trace. dual of DD case §3.13 under the up/down bridge ($BB \sim DD$).

Case $a_1, a_2 \neq 0$.



Trace. dual of DD case §3.13 under the up/down bridge ($BB \sim DD$): the dual of the [CZ02-CZ-CZ] core.